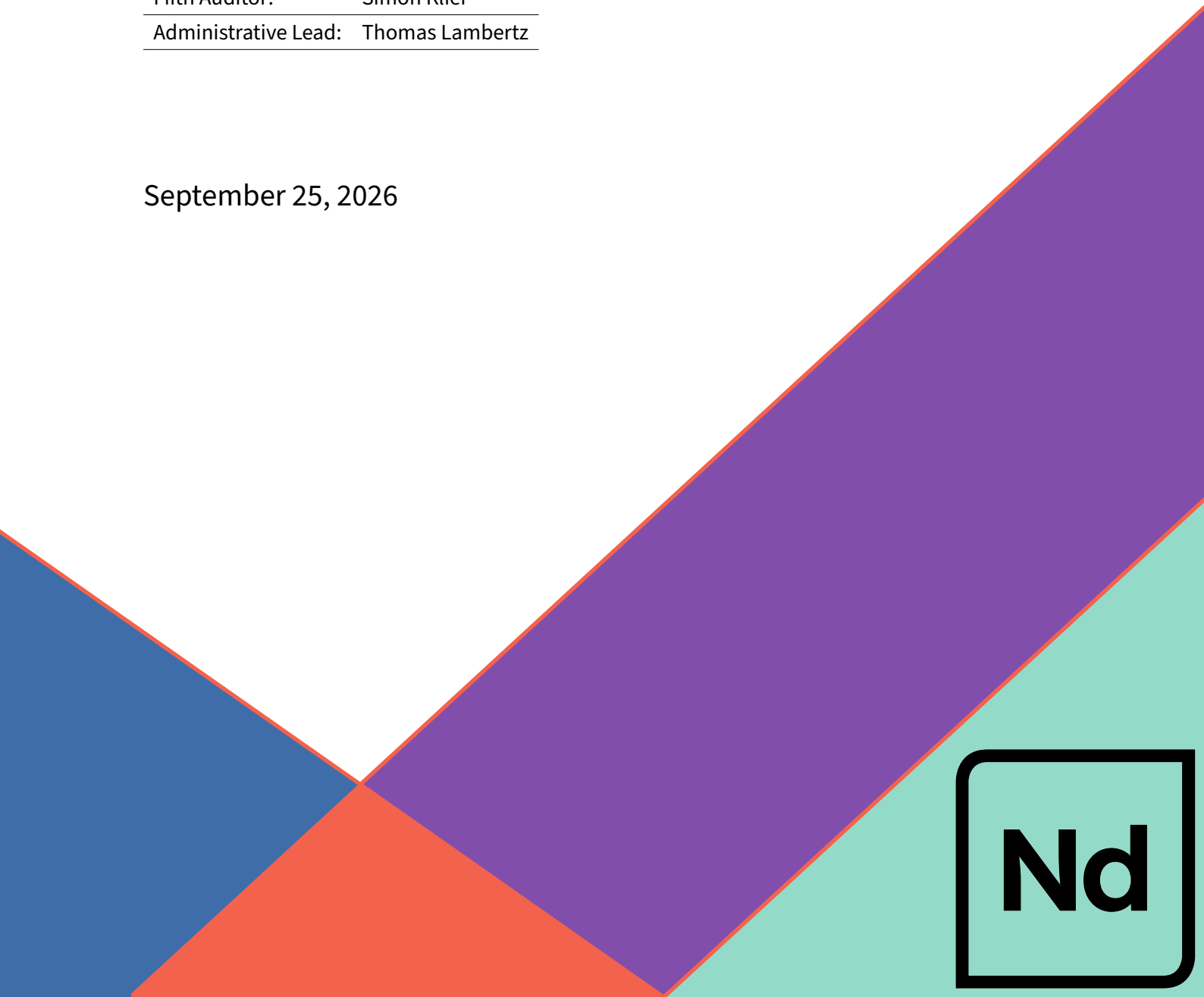

Point Release Audit - Agave v4.3

conducted by Neodyme AG

Lead Auditor:	Thomas Lambertz
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Third Auditor:	Justin Mietzner
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Fifth Auditor:	Simon Klier
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September 25, 2026



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1 | Executive Summary

On September 25, 2026, Neodyme completed a Point Release Audit of Anza's Agave changes between the v4.2.0 tag (ac82b5d438) and the v4.3.0 tag (825efd1829). The review covered five assigned functional areas. The Alpenglow consensus migration was audited separately and was not part of this audit.

Neodyme identified no security-relevant findings during this audit. In particular, no critical- or high-severity issues were found.

2 | Scope

Reviewed Revision

Boundary	Reference	Commit
Start	v4.2.0	ac82b5d438
End	v4.3.0	825efd1829

Neodyme reviewed changes to:

- **Transaction Ingestion Pipeline:** QUIC peer admission and stake revalidation, transaction-client and forwarding changes, and packet handling during signature verification.
- **Leader Schedule & Block Production:** leader-schedule publication, PoH recording and pacing, fast leader handover and replay, and block-marker validation.
- **Block Propagation (Turbine/Shreds/XDP):** Turbine broadcast and shred validation, block-store replay and cleanup, repair-request and proof validation, and the XDP transmission path.
- **Accounts DB & State Storage:** account indexing and deleted-account handling, lattice-hash calculation, and snapshot creation and restoration. This included the AVX2- and AVX-512-optimized BLAKE3 implementation and its integration into startup index generation ([PR #13595](#)).
- **Gossip & P2P Communication:** Ping/Pong peer validation and cache limits, pull-request scan accounting, CRDS data retention and filtering, and notifications sent to plugins.

Exclusions

The Alpenglow consensus migration, areas assigned to OtterSec, and non-production components such as CI, scripts, CLI, RPC, tests, and benchmarks were outside Neodyme's scope. Changes in the areas listed above remained in scope where they interacted with Alpenglow.

3 | Findings

Neodyme identified no security-relevant findings during this audit. In particular, no critical- or high-severity issues were found.

4 | About Neodyme

Security is difficult.

To understand and break complex things, you need a certain type of people: people who thrive in complexity, who love to play around with code, and who do not stop exploring until they fully understand every aspect of it. That's us.

Our team never outsources audits. Having found over 80 high- or critical-severity bugs in Solana's core code itself, we believe that Neodyme hosts the most qualified auditors for Solana programs. We have also found and disclosed critical vulnerabilities in many of Solana's top projects and responsibly disclosed issues that could have resulted in the theft of over \$10B in TVL on the Solana blockchain.

All of our team members have a background in competitive hacking. During hacking competitions known as CTFs, we competed and collaborated while finding vulnerabilities, breaking encryption, reverse engineering complex algorithms, and much more. Many of our team members have won national and international competitions and continue to rank highly in some of the world's hardest CTF events.

In 2020, some of our members started experimenting with validators and became active members of the early Solana community. After reporting several bugs, the Solana Foundation contracted us for source-code auditing in 2021. As a result, Neodyme was born.

